

Ray Foysal

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Education

Australian National University (ANU) | Canberra, Australia

2025 - Present

- Completing 3rd year of mechanical engineering at ANU as part of an Exchange Programme

University College Dublin (UCD) | Dublin, Ireland

2023 - Present

- Grade 1:1
- Relevant Modules: Mechanics for Engineers, Mechanics of Fluids, Manufacturing Engineering, Energy Engineering, Physics for Engineers, Computer Science for Engineers, Computer Engineering, Robotics Design Project, Electrical and Electronic Circuits, Creativity in Design.

Colaiste Pobail Setanta | Dublin, Ireland

2018 - 2023

- Leaving Certificate Points Obtained: **613**

Best Senior Certificate Award | Colaiste Pobail Setanta

June 2024

- Awarded to the senior from secondary school with the highest leaving certificate result.

Experience

UCD UAV | Aeromechanical & Simulations Engineer

Feb 2024 – Jul 2025

- Utilised X-Plane 11 simulations to refine flight characteristics of different planes in our prototype series. Scored 128/150 points in iMechE competition on first attempt.
- Used SimScale to simulate and verify properties of potential landing gear components
- Led the design and construction of a custom built ground control station in conjunction with the avionics department. Stand out features include deployable table to set up and operate the UAV.
- Conducted analysis of air foil designs and lift/drag properties using XFLR5 CFD software.
- Won 1st place in airworthiness and business presentation in Buckminster UK, while achieving runner-up in 4 other categories out of 8 total against 32 international teams.

UCD Formula Student | Powertrain Engineer

Sep 2023 – Present

- Designed and developed the High Voltage Battery Enclosure using Onshape.
- Evaluated appropriate high voltage fuses for use in internal battery connections, and created technical reports for review by faculty advisors.
- Used KiCAD to reference circuit board schematics to solder components correctly.
- Assembled wheel connections such as tie rods, knuckles, axle, and suspension components.
- Successfully passed battery scrutineering in the IMechE FSUK 2025 competition as a member of the UCDFS team.
- Won 3rd place out of 91 for business presentation, and 28th position overall out of 59 teams at Silverstone.

Work Experience

Red Cow Moran Hotel | F&B Assistant

Sep 2023 – April 2025

- Received Manual Handling and HACCP training.
- Worked in a fast paced environment, served 100+ people every day.

Talbot Hotel Stillorgan | F&B Assistant / Receptionist

February – March 2022

- Assisted in daily operations, managed guest inquiries, and helped to ensure a high level of 95% customer satisfaction.

Technical Skills

Programming

- Proficiency of C/C++ Programming from Computer Science, Computer Engineering and Robotics modules undertaken at UCD.
- Proficiency of Python from Programming for Scientists, and Environmental Sensing and Mapping modules taken at ANU.

Modelling/Simulating

- Proficiency in SOLIDWORKS, OnShape and Autodesk Fusion 360
- Knowledge of MATLAB and Simulink from Engineering Computing course undertaken in UCD, and from battery simulation trials for the formula student team in UCD.
- Knowledge of Blender 3D modelling software to create several animations for Engineers Without Borders video presentation and for product advertisements for personal e-commerce business.
- Knowledge of Simulating software such as SimScale, XFLR5, and X-Plane 11 from experience working on the UCD UAV team.

Manufacturing

- Knowledge in 3D printing, manual machining and CNC machining.
- Knowledge of materials and associated properties; thermoset/thermoplastic polymers, composites, metals etc.

Circuitry & Electronics

- Knowledge in LTSpice simulation and KiCAD PCB Modelling.
- Knowledge in Arduino, sensors, shields, and circuitry from multiple projects; RoboWars, Robotics Design Module, MicroMouse Competition, NASA SpaceApps Hackathon.

Microsoft Suite

- Proficiency in MS Excel from specific coursework done in Intro to Eng. Computing, along with MS Word and PowerPoint.

Achievements/Extra Curricular

Dogpatch RoboWars 2025 | Winner

July 2025

- Competed in RoboWars battle organised by Dogpatch, a startup and innovation hub in Dublin's docklands known as the city's "Silicon Docks".
- Worked in a team of 6 people to create a robot from scratch to fight in the timeframe of 1 month.

NASA Space Apps Hackathon | Winner

October 2023

- Achieved first place in the largest hackathon in Ireland, using NASA's open data.
- Responsible for designing and 3d printing the prototype beacon that detects forest fires from its initial stages and warns local authorities.

Produced AI Podcast

February 2024

- Served as the producer for a podcast episode on AI and its impact on reflective assignments as part of a series from the UCD College of Engineering and Architecture.
- Collaborated with the facilitators, team members and guests to plan, script and record the episode.
- Coordinated logistics, including managing recording equipment, booking recording times at a studio and post-production editing to ensure high quality content delivery.

UCD Entrance Scholar

March 2024

- Awarded UCD Entrance Scholar Award for achieving outstanding result in the Leaving Certificate and graduating as the top student in secondary school.

Robotics Design Competition | Software Developer and Design Engineer

May 2024

- Developed and implemented a strategy for the robot in C programming running on Arduino.
- Designed mechanical systems within the robot to combat and outlast opponents, including drivetrain, ball-capturing mechanism and obstacle placement system.

Sports Star of The Year

June 2021

- Recognised as the best performing student athlete in secondary school.